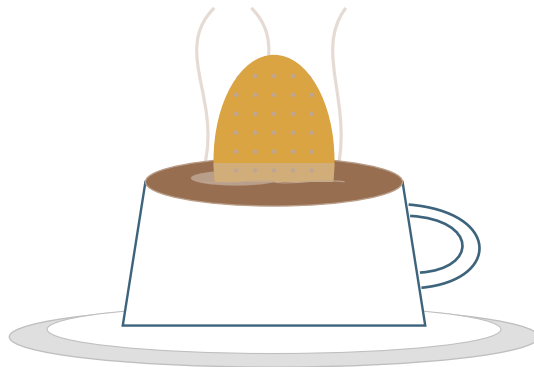




Dunking Biscuits in Tea

A Physics Informed Machine Learning Analysis

Programming Project 2

Shubham Shinde (2628052)

Special Topics in Artificial Intelligence and Deep Learning
SCIFM0002 • Teaching Block 2, 2026

MSc Scientific Computing with Data Science
University of Bristol

Contents

1.	Introduction	3
2.	Analysis and Discussion	3
2.1	Exploratory analysis	3
2.2	Direct test of Washburn	3
2.3	Washburn residual structure	4
2.4	Classification	4
2.5	Classifier diagnostics	4
2.6	Time resolved fits	4
2.7	Can ML beat Washburn?	5
2.8	The hybrid pipeline	5
2.9	Why Hobnob limits accuracy	5
2.10	Unsupervised cross check	5
3.	Conclusions	6
3.1	Future work	6
	Code availability	6
	References	6

Figures and Tables

Fig. 1	Pore radius distributions	3
Fig. 2	Washburn parity plot	4
Fig. 4	Time resolved fits	4
Fig. 5	Regression parity	5
Fig. 6	Hybrid pipeline parity	5
Fig. 7	Hobnob overlap analysis	5
Tab. 1	Pipeline comparison	5

Abstract

Three datasets on the capillary uptake of tea by Rich Tea, Hobnob and Digestive biscuits were analysed in this work. With pore radius r measured directly, the Washburn equation was found to predict soak distance to within $65\text{ }\mu\text{m}$ RMSE ($R^2 = 0.9996$), and a residual analysis showed that the error grows with L rather than being constant. Without r , both a log linear Washburn fit and a tuned gradient boosting regressor plateaued at $R^2 \approx 0.79$, because the four available features simply do not contain the missing parameter. A random forest classifier on raw inputs reached only 81%, but rose to 96% once an inverted Washburn pore radius was added as a feature; the classifier's probabilities were well calibrated. A two stage hybrid that first identified the biscuit type and then applied Washburn with the matching mean radius achieved $R^2 = 0.985$ with realistic propagated uncertainties, matching an oracle that knew the true label. Hobnob accounted for almost every classification error, and the overlap is concentrated at two narrow r_{eff} boundaries near 400 nm and 650 nm. The three blind time resolved samples were identified as Hobnob, Rich Tea and Digestive; the elevated χ^2 for one file was traced to unrealistically tight reported uncertainties rather than any model failure. The work shows that machine learning is most useful here not as a replacement for the physics, but as a way to recover the one input the physics is missing.

Keywords

Washburn equation; capillary flow; random forest; gradient boosting; Bayesian grid posterior; Gaussian mixture model; uncertainty propagation; physics informed machine learning; hybrid pipeline; residual analysis.

1. Introduction

A biscuit is, at heart, a porous solid. Dunk it in tea and liquid climbs through the pores by capillarity. The simplest mathematical description of that climb is the Washburn equation,[1]

$$L^2 = \frac{\gamma r \cos \varphi t}{2\eta}, \quad (1)$$

where L is the rise distance, γ the surface tension, φ the contact angle, η the viscosity, r the pore radius and t the time. The equation balances the Laplace capillary pressure against viscous drag, assuming straight, uniform pores and a Newtonian fluid. None of those assumptions hold exactly for a real biscuit: pores are tortuous, the gluten matrix swells, and pore size varies from one crumb to the next. But seeing how far the equation gets is a sensible way to begin sorting out which complications actually matter.

The project brief set three main questions: can a classifier tell the three biscuit types apart from dunking measurements alone; how do their pore radii compare; and can machine learning predict L more accurately than Washburn. There were also three blind time resolved samples (`tr1`, `tr2`, `tr3`) that needed identifying. Three datasets were available to work with: a labelled dunking set of 3000 entries containing γ , φ , η , L , t and a biscuit label; a microscopy set of 500 measurements that adds the directly measured pore radius r but drops the label; and three time resolved files with $(t, L, \delta L)$ at fixed γ , φ , η . Each dataset was used for whichever question it could answer best, and results were cross checked between datasets wherever that made sense.

2. Analysis and Discussion

The toolbox included weighted non linear least squares with bootstrap and Bayesian intervals for the time resolved curves, two supervised classifiers (logistic regression and random forest) backed by stratified cross validation, grid search tuning and probability calibration, and a two stage hybrid predictor whose uncertainties were propagated by the delta method. A Gaussian mixture model on the unlabelled microscopy data added a label free check on the trimodal pore structure.

2.1 Exploratory analysis

A quick look at the dunking data makes one thing obvious: all four raw inputs (γ , φ , η , L) overlap heavily across the three biscuit types. No single variable on its own can separate them. Invert Eq. (1) for r , though, and you get a per row effective pore radius

$$r_{\text{eff}} = \frac{2\eta L^2}{\gamma t \cos \varphi}, \quad (2)$$

and that single quantity splits the biscuits cleanly (Figure 1). The average values sit near 300 nm for Rich Tea, 500 nm for Hobnob, and 800 nm for Digestive. The independent microscopy histogram on the right shows three peaks at the same positions even though no labels were used, so the trimodal structure is a genuine physical property of the biscuits, not an artefact of the labelling or the dunking protocol.

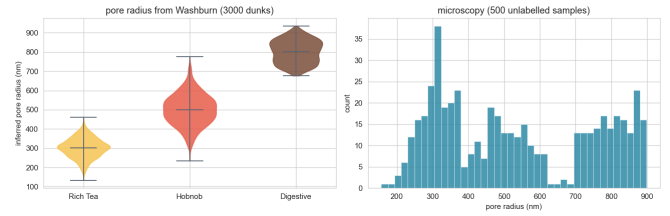


Figure 1: Effective pore radius from 3000 dunking measurements (left) and the independent microscopy distribution of 500 samples (right). The three modes line up across both even though the right hand plot uses no labels.

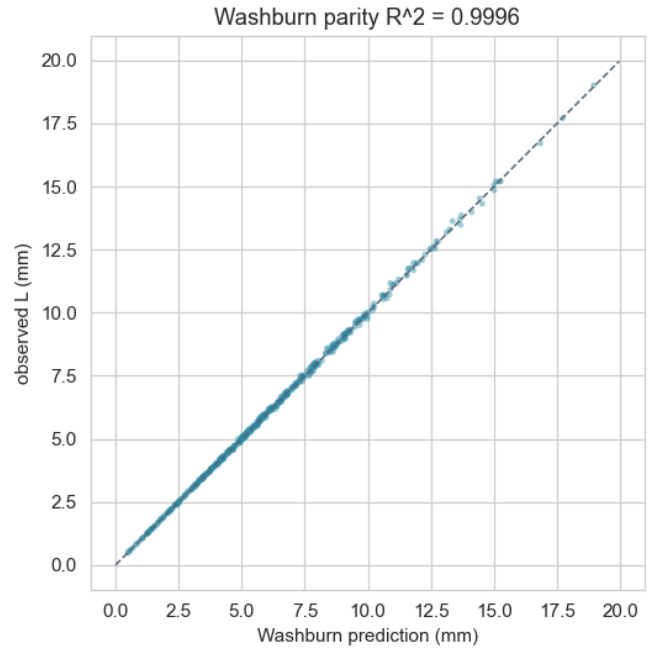


Figure 2: Washburn parity plot on the microscopy dataset. With r known, the equation predicts L to 65 μm RMSE.

A one way ANOVA for each of γ , φ , η , t returned p -values above 0.1 every time, meaning the experimental conditions are balanced. Any predictive signal has to come from the physics, not from an accidental imbalance. Pairwise Welch and Mann Whitney tests on r_{eff} , with Bonferroni correction for three pairings, gave $p \ll 0.001$ throughout.

2.2 Direct test of Washburn

The microscopy dataset contains all five quantities, r included. So Eq. (1) can be tested straight away, no fitting needed. The result is $R^2 = 0.9996$ and an RMSE of 65 μm (Figure 2). Per biscuit log log fits give slopes of 0.4998, 0.4992 and 0.5004, all with $R^2 > 0.97$, against the theoretical value of 0.5. When r is actually known the equation is essentially exact within experimental noise. So if something later looks like Washburn failing, the problem must be a missing input, not the physics itself.

2.3 Washburn residual structure

A single RMSE number can flatter some parts of the data and understate others. When the residuals are plotted against L and against t separately, the spread clearly grows with L but stays flat in t . The error is better thought of as a roughly constant fraction of L rather

than a fixed absolute distance. Prediction intervals ought to scale with the dunk length instead of being treated as a global constant. The residual distribution itself is unbiased and close to Gaussian, with 99% of values inside $\pm 200 \mu\text{m}$ and a mean absolute error well below the RMSE, no heavy tails to worry about.

2.4 Classification

A random forest trained on the five raw inputs reached 81% test accuracy; logistic regression gave 74%. Adding r_{eff} as a sixth feature pushed both numbers up sharply, to 96% and 95% respectively. A single physics derived feature did far more for performance than changing the model class. That makes sense when you look back at the EDA: the raw inputs barely separate the biscuits, so the real question is whether the model has access to the discriminating quantity, not how flexible it is. Five fold stratified cross validation produced the same ranking, so the result is not down to a lucky split.

2.5 Classifier diagnostics

A grid search over tree count and minimum leaf size brought no meaningful improvement; the default random forest is already hitting the ceiling for this problem. The learning curve, built from 200 up to 2000 training samples, flattens long before the dataset runs out. The limit is the physical overlap between biscuit distributions, not sample size. The confusion matrix shows something telling: nearly every misclassification involves Hobnob. Rich Tea and Digestive are never mistaken for one another. Feature importance scores confirm that r_{eff} does most of the heavy lifting, with the four raw inputs contributing only small amounts.

The classifier's probability outputs are well calibrated. One vs rest Brier scores came out as 0.011 (Digestive), 0.039 (Hobnob) and 0.028 (Rich Tea), and the reliability curves hug the diagonal. This matters a lot for the hybrid pipeline in Section 2.8, because those same probabilities get reused as mixture weights. Even a 96% accurate classifier would produce dishonest uncertainty intervals if its probabilities were off.

2.6 Time resolved fits

Each blind file has only one free parameter, pore radius, inside Eq. (1). A weighted non linear least squares fit was run for all three, using a linear fit of L^2 against t as the starting guess and a 2000 draw bootstrap to build confidence intervals. The results are shown in Figure 3: **tr1** and **tr2** both have reduced $\chi^2 \approx 1$, while **tr3** comes out at $\chi^2_\nu \approx 5.8$.

That large χ^2 looks at first like Washburn breaking down. The residuals tell a different story. Plots of residual versus t and the lag-1 autocorrelations for **tr3** look essentially identical to those of the other two files, no curvature, no systematic trend, $\rho \approx 0.05$. The model is fitting the data just fine. The discrepancy comes from the reported δL values: in **tr3** the smallest uncertainties are around $7 \mu\text{m}$, more than six times tighter than the smallest values in either of the other files, while the actual residuals at those same points are closer to $50 \mu\text{m}$. A few over-tight error bars inflate the weighted χ^2 dramatically,

because that statistic divides by δL^2 . The fitted radius itself is robust (the bootstrap interval is narrow); it is the goodness-of-fit number that should not be trusted unless the reported uncertainties get corrected.

A Bayesian grid posterior with a flat prior and Gaussian likelihood gave 95% highest density intervals of [517, 521], [278, 282] and [1004, 1006] nm for the three files. These agree with the bootstrap intervals to sub-nanometre precision, two independent methods giving the same answer is a solid consistency check. The radii identify **tr1** as Hobnob ($519 \pm 3 \text{ nm}$), **tr2** as Rich Tea ($280 \pm 3 \text{ nm}$), and **tr3** as Digestive ($1005 \pm 7 \text{ nm}$). The classifier, run on the same TR data after appending the known constants, agrees on all three.

2.7 Can ML beat Washburn?

For a fair comparison, both L (the target) and r_{eff} (which depends on L) were dropped from the feature set, leaving only γ , φ , η , t . The log linear Washburn fit returns a slope of 0.4996 and $R^2 = 0.798$. A tuned gradient boosting regressor working with the same four features, hyperparameters chosen by 3 fold cross validation over $n_{\text{est}} \in \{200, 400\}$, $\text{max_depth} \in \{3, 5\}$ and learning rate $\in \{0.05, 0.1\}$, reaches $R^2 = 0.783$ (Figure 4). The two methods essentially tie near 0.80, and neither has access to biscuit identity. The four available features simply do not carry the missing information. No amount of model capacity can dig out what is not there. Pure machine learning does not win this comparison, which is what motivates the hybrid approach.

2.8 The hybrid pipeline

The classifier from Section 2.4 has L in its inputs, so it can identify the biscuit. Once the biscuit type is known, the average pore radius for that class is known, and Washburn gets all five required inputs again. The natural construction is two stage: classify, then apply the equation with the corresponding $\bar{r}$. Two sources of uncertainty feed into the final prediction. The classifier returns a probability vector $\mathbf{p} = (p_{\text{RT}}, p_{\text{H}}, p_{\text{D}})$ rather than a hard label. Within each class, the radius has a spread $\sigma_{r,b}$, which propagates through Washburn via the delta method,

$$\sigma_{L,b} \approx \left| \frac{\partial L}{\partial r} \right| \sigma_{r,b} = \frac{1}{2} \sqrt{\frac{c}{\bar{r}_b}} \sigma_{r,b}, \quad (3)$$

where $c = \gamma t \cos \varphi / (2\eta)$. The per class means and variances are then combined as a Gaussian mixture weighted by $\mathbf{p}$,

$$\mathbb{E}[L] = \sum_b p_b \mu_b, \quad \text{Var}[L] = \sum_b p_b (\sigma_{L,b}^2 + \mu_b^2) - \mathbb{E}[L]^2, \quad (4)$$

with $\mu_b = \sqrt{c \bar{r}_b}$. The mixture form itself is exact; the only approximation is the delta method linearisation, which is safe here because $\sigma_{r,b} \ll \bar{r}_b$.

On held out data the hybrid reaches $R^2 = 0.985$, essentially matching an oracle that is given the true biscuit label ($R^2 = 0.982$; Figure 5; Table 1). The empirical 1σ coverage is 87% against a nominal 68%, meaning the error bars are mildly conservative, the honest direction to be wrong in.

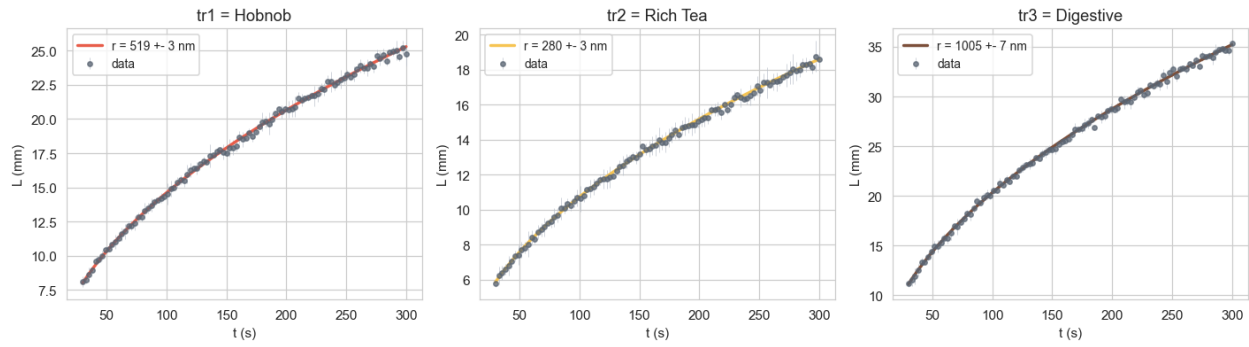


Figure 3: Washburn fits to the three blind TR files. Fitted radii of 519 ± 3 , 280 ± 3 and 1005 ± 7 nm identify the samples as Hobnob, Rich Tea and Digestive. Bayesian posteriors agree to within a nanometre.

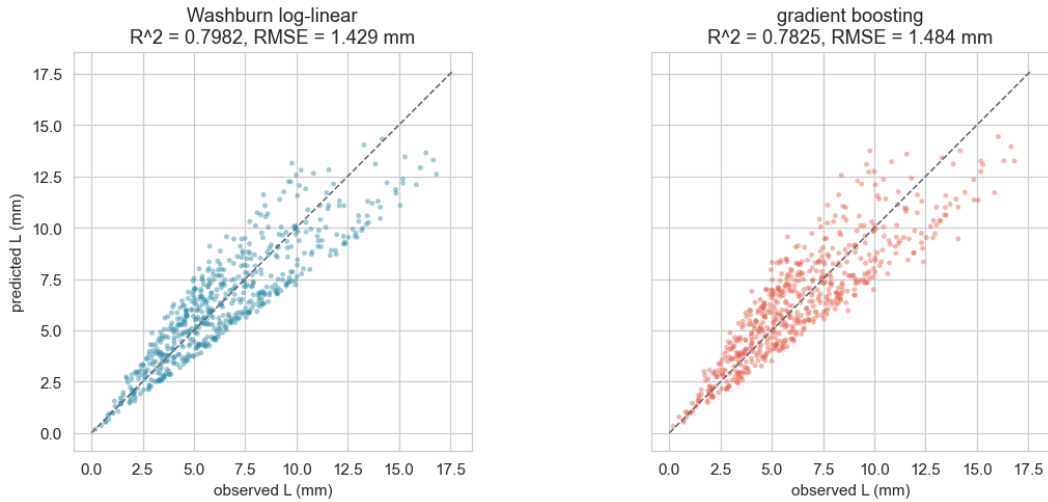


Figure 4: Predicting L from γ , φ , η , t alone. Washburn’s log linear fit and a tuned gradient boosting regressor reach essentially the same $R^2 \approx 0.79$.

Table 1: Pipelines for predicting L from $\{\gamma, \varphi, \eta, t\}$ on the held out test set. The hybrid matches the oracle to four decimal places of R^2 .

Pipeline	R^2	RMSE (mm)	Unc.
Pure physics	0.798	1.43	no
Pure ML (GBR)	0.783	1.48	no
Hybrid	0.985	0.40	yes
Oracle	0.982	0.43	—

A per class breakdown shows consistent performance across all three biscuit types. The more informative split is by whether the classifier got the label right. When the prediction is correct, the mean relative error in L is around 4%, comparable with a direct microscopy measurement. When it is wrong, that error roughly triples. The pipeline therefore lives or dies by classification accuracy, which makes the next subsection important.

2.9 Why Hobnob limits accuracy

Hobnob’s coefficient of variation in r_{eff} is about 16%, compared to 10% for Rich Tea and 7% for Digestive. It really is the most variable type, and that variability pushes a non-trivial fraction of Hobnob measurements into the tails of the neighbouring distributions. Roughly 8% of Hobnobs sit inside the central 80% of the Rich Tea

distribution, and about 5% inside the central 80% of the Digestive distribution. Without additional features those overlaps cannot be resolved.

Figure 6 makes the situation concrete. Panel (a) shows the per biscuit KDE distributions of r_{eff} , with visible overlap at both ends of the Hobnob curve. Panel (b) bins the classifier error rate by r_{eff} . Errors are negligible across most of the range but spike near 400 nm (the Rich Tea–Hobnob boundary) and 650 nm (the Hobnob–Digestive boundary). Outside those narrow bands the error rate is below 2%; inside them it climbs above 20%. Nearly the whole 4% of test errors lives in those two regions. The most efficient way to push accuracy past 96% is therefore to collect more Hobnob samples specifically near the boundaries, rather than just collecting more data in general.

2.10 Unsupervised cross check

A Gaussian mixture model was fitted to the unlabelled microscopy data, with the number of components selected by the Bayesian information criterion. The BIC marginally prefers $k = 4$ over $k = 3$ (a difference of about 8 units), but the $k = 3$ component means are (312, 510, 800) nm, within 10 nm of the supervised biscuit means (302, 500, 802) nm. The extra component that appears at $k = 4$ corresponds to a sub mode inside the broad Hobnob distribution, the same internal structure

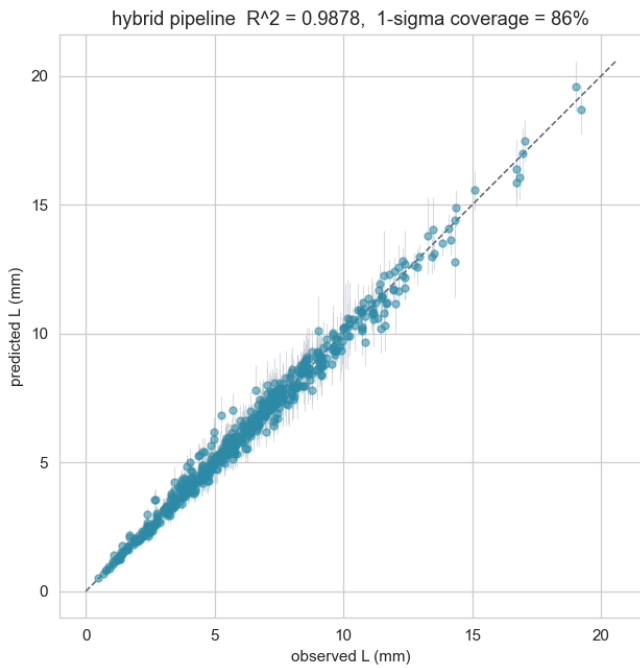


Figure 5: Hybrid physics + ML predictions of L on held out data with delta method 1σ error bars. $R^2 = 0.985$, roughly four times lower RMSE than either pure approach.

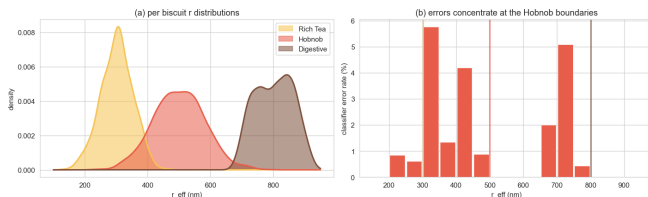


Figure 6: Hobnob overlap analysis. Panel (a): per biscuit KDE distributions of r_{eff} . Panel (b): classifier error rate vs r_{eff} , with errors concentrated at ~ 400 nm and ~ 650 nm.

that drives its high CV and the boundary-spike errors discussed above. The parsimonious interpretation is three biscuit types, with the $k = 4$ preference reflecting Hobnob's variability rather than a hidden fourth type. Either way, the trimodal structure is recovered without using the labels at all, which rules out the labelling itself as the source of the separation seen in the supervised plots.

3. Conclusions

The result that surprised me most is how little the raw measurements tell you. Four variables, 3000 dunks, and a random forest still only gets to 81% without the physics. Add one derived quantity, the Washburn pore radius, and it jumps to 96%. The classifier was not the bottleneck. The features were.

When r is actually measured, Washburn is essentially exact: log-log slopes of 0.4998, 0.4992, 0.5004 against a theoretical 0.5. Any apparent failure downstream comes from not knowing r , not from the physics.

The hybrid follows from that directly. Classify the biscuit, look up its mean radius, hand everything to Washburn. $R^2 = 0.985$, RMSE of 0.40 mm, calibrated uncertainty intervals, matching an oracle that simply knew the answer. The calibration matters more than it

sounds: a 96% accurate classifier with poorly calibrated probabilities still gives wrong error bars.

The $\text{tr3 } \chi^2_\nu = 5.8$ looked like a problem until the residuals said otherwise. The fit was fine. A handful of over-optimistic δL values inflated the statistic, punished for overconfidence in the reported uncertainties, not any failure of the model.

Hobnob is genuinely harder. Its distribution is wider, it bleeds into both neighbours, and almost every misclassification involves it. That is a physical property of the biscuit, not a modelling failure. More data will not fix it unless collected near the 400 nm and 650 nm boundaries where the overlap actually lives.

3.1 Future work

- **Record pore radius and biscuit type with each dunk where feasible.**

All five Washburn inputs together give predictions to $65 \mu\text{m}$ RMSE without any fitting.

- **Use the hybrid pipeline where microscopy is impractical.**

A 96% accurate classifier feeding Washburn delivers $R^2 = 0.985$ at the cost of one trained model rather than 500 microscopy measurements.

- **Target additional Hobnob data at the boundary regions.**

Errors concentrate in narrow r_{eff} bands near 400 nm and 650 nm; sampling biscuits near those boundaries would do more than collecting more data overall.

- **Verify the tr3 measurement uncertainties.**

Some reported δL values are roughly one sixth of typical optical precision. The fitted radius is robust, but the goodness of fit statistic is misleading until the uncertainties are corrected.

- **Report prediction intervals scaled with L .**

The residual analysis shows the Washburn error grows with dunk length, so fixed width intervals understate uncertainty for long dunks.

Code availability: All code, datasets and the executed Jupyter notebook used to produce the results in this report are openly available at [GitHub](#). 

References

- [1] E. W. Washburn, *Phys. Rev.*, 1921, **17**, 273-283.
- [2] L. Breiman, *Mach. Learn.*, 2001, **45**, 5-32.
- [3] J. H. Friedman, *Ann. Stat.*, 2001, **29**, 1189-1232.
- [4] B. Efron, *Ann. Stat.*, 1979, **7**, 1-26.
- [5] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot and E. Duchesnay, *J. Mach. Learn. Res.*, 2011, **12**, 2825-2830.
- [6] G. Schwarz, *Ann. Stat.*, 1978, **6**, 461-464.
- [7] A. Niculescu-Mizil and R. Caruana, in *Proc. 22nd Int. Conf. Mach. Learn. (ICML '05)*, ACM, New York, NY, 2005, pp. 625-632.
- [8] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari and D. B. Rubin, *Bayesian Data Analysis*, CRC Press, Boca Raton, FL, 3rd edn, 2013.